CLASS 8 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 1–4: **Exploring the Investigative World of Science, Invisible Living World: Beyond Our Naked Eye, Health: The Ultimate Treasure, Electricity: Magnetic and Heating Effects** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A health camp checks weight, height and haemoglobin. This is: 1
(a) vaccination **(b)** mental health therapy
(c) monitoring physical health **(d)** treatment of an infection

2. A student connects two cells in series in a circuit with a bulb. The bulb: 1
(a) goes out **(b)** glows dimmer **(c)** does not glow **(d)** glows brighter

3. To test whether the type of soil affects plant growth, which factor should be varied? 1
(a) amount of water **(b)** seed type **(c)** amount of light **(d)** type of soil

4. Which two are found in both a plant and an animal cell? 1
(a) wall and vacuole **(b)** nucleus and cytoplasm
(c) chloroplast and wall **(d)** chloroplast and nucleus only

5. Mucus in the nose helps by: 1
(a) producing vitamins **(b)** trapping dust and germs
(c) digesting food **(d)** making noise

6. A wire carrying current produces a field in the form of: 1
(a) circles around the wire **(b)** straight lines along the wire
(c) dots **(d)** no pattern

7. If 20 swings take 36 s, the time period of one swing is: 1
(a) 0.18 s **(b)** 720 s **(c)** 18 s **(d)** 1.8 s

8. Which of these statements about bacteria is correct? 1
(a) They are visible to the naked eye.
(b) They have a nucleus and chloroplasts.
(c) They have genetic material but no true nucleus.
(d) They are always harmful.

9. A student recovered from chickenpox and does not get it again. This is due to: 1
(a) exercise **(b)** diet **(c)** fever **(d)** immunity

10. An electromagnet lifts 10 clips with 20 turns. With 40 turns of the same wire and cell, it will lift: 1
(a) more clips **(b)** the same number of clips
(c) no clips **(d)** fewer clips

11. An observation that cannot be repeated by others is: 1

(a) proven **(b)** unreliable **(c)** reliable **(d)** a law

12. A scientist sees under a microscope a tiny moving single-celled organism with a nucleus in pond water. It is likely to be a: 1

(a) bacterium **(b)** mushroom **(c)** virus **(d)** protozoan

13. Which is a lifestyle disease? 1

(a) measles **(b)** type 2 diabetes **(c)** chickenpox **(d)** cholera

14. A thick copper wire is chosen for home wiring because it: 1

(a) is an insulator **(b)** melts easily

(c) heats less due to low resistance **(d)** is a magnet

15. Which question needs measurement to be answered? 1

(a) Is ice cold feeling good?

(b) Is ice pretty?

(c) Do you like ice?

(d) How long does ice take to melt in a warm room?

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do the doctors give oral rehydration solution (ORS) in diarrhoea? 2

17. A thin and thick nichrome wire of equal length have the same current. Which heats more and why? 2

18. Calculate the mean of the readings 4.2 s, 4.0 s, 4.1 s and 4.3 s. 2

19. An eyepiece of 15x and an objective of 20x are used. Find the magnification. 2

20. Why must a patient with an infectious disease like measles be kept away from other children? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A village had cases of cholera. List three public health steps to stop it. 3

22. Explain why long thin wires are not used for the household wiring but short thick wires with insulation are. 3

23. Why must the experimenter note the units of each quantity in a table? 3

24. Name the organelles responsible for 3

(a) photosynthesis

(b) control

(c) storage in plant cells.

OR

How do physical, mental and social aspects of health influence each other? Give an example. 3

25. Why are electric appliances fitted with a three-pin plug having an earth pin? State the role of earth. 3

OR

Explain why many experiments need a time limit and fixed procedure. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Marks of a pendulum experiment: length (cm) 20, 40, 60 and time period (s) 0.9, 1.3, 1.6. Describe the relationship and a conclusion. State two variables to keep constant. 5

OR

Explain the significance of a microscope in biology. Give the parts, the working and the calculation of magnification for 12x eyepiece and 50x objective. 5

27. Describe the working of the immune system against pathogens, including the roles of skin, white blood cells and antibodies. 5

OR

Explain the construction and working of an electric bell with a flow chart in words. 5

END OF QUESTION PAPER — **Total: 50 Marks**

MASTERMIND